第一章：场论及预备知识

电磁场与电磁波

场论及预备知识

1. Hamilton 算符 $\nabla=\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right)=\frac{\partial}{\partial x}\vec{i}+\frac{\partial}{\partial y}\vec{j}+\frac{\partial}{\partial z}\vec{k}$

∇ 读作 Nabla

2. 由 Hamilton 算符得到梯度、散度旋度的定义：

设 $u=u(x,y,z)$，$A=P(x,y,z)\vec{i}+Q(x,y,z)\vec{j}+R(x,y,z)\vec{k}$

u 与 A 都为光滑数量场，P、Q、R 分别为 A 在 $\vec{i}$、$\vec{j}$、$\vec{k}$ 方向上的分量。则，由 u 和 A 产生的梯度、旋度、散度场为

$$\text{grad}\ u=\nabla u=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)=\frac{\partial u}{\partial x}\vec{i}+\frac{\partial u}{\partial y}\vec{j}+\frac{\partial u}{\partial z}\vec{k}$$

梯度为最大方向导数

散度 $\text{div}\ u=\nabla\cdot A=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=\lim_{\Delta V\to 0}\frac{1}{\Delta V}\oint_S A\cdot dS$

散度为通量的体密度

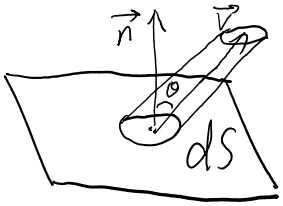
旋度 $\text{rot}\ A=\nabla\times A=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ P&Q&R\end{vmatrix}=\left(\frac{\partial R}{\partial y}-\frac{\partial Q}{\partial z},\frac{\partial P}{\partial z}-\frac{\partial R}{\partial x},\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)$

$$=\left(\frac{\partial R}{\partial y}-\frac{\partial Q}{\partial z}\right)\vec{i}+\left(\frac{\partial P}{\partial z}-\frac{\partial R}{\partial x}\right)\vec{j}+\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)\vec{k}$$

旋度为环量的面密度

补充：环量与通量：环量：$\Gamma=\oint_L \vec{A}\cdot d\vec{l}$

Flux 通量 矢量 E 沿有向曲面 S 的面积分



$$\Phi=\int_S E\cdot dS$$

$$dQ=ds\ \vec{n}=ds\,|\vec{v}|\cdot\cos\theta=\vec{v}\cdot d\vec{S}$$

$$Q=\int_S dQ=\int_S \vec{v}\cdot d\vec{S}$$

$\nabla\cdot(c\vec{A})=c\,\nabla\vec{A}$　　结论：① $\nabla\times(\nabla u)=0$ 梯度无散

$\nabla\cdot(c\vec{A}) = c\nabla\vec{A}$

$\nabla(\vec{A}+\vec{B}) = \nabla\cdot\vec{A} + \nabla\cdot\vec{B}$

$\nabla(u\vec{A}) = \nabla\cdot u\cdot\vec{A} + u(\nabla\cdot\vec{A})$

结论：① $\nabla\times(\nabla u) = 0$ 梯度无散

② $\nabla\cdot(\nabla\times\vec{A}) = 0$ 旋度无散

③ $\nabla\times(u\vec{A}) = u\nabla\times\vec{A} + \nabla u\times\vec{A}$

④ $\nabla(\vec{A}\times\vec{B}) = \vec{B}\cdot\nabla\times\vec{A} - \vec{A}\nabla\times\vec{B}$

3. 矢量混合积：$A\cdot(B\times C) = B\cdot(C\times A) = C\cdot(A\times B)$

4. Laplacian 算子 $\nabla^2 = \nabla\cdot\nabla = \frac{\partial^2}{\partial x} + \frac{\partial^2}{\partial y} + \frac{\partial^2}{\partial z}$

对于标量函数：$\nabla^2\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2}$

对于矢量函数：$\nabla^2 A = \frac{\partial^2 A}{\partial x} + \frac{\partial^2 A}{\partial y} + \frac{\partial^2 A}{\partial z}$